

DAY 10

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JupyterLab Python 3 (ipykernel)
```

```
[5]: import numpy as np

#dataset
x=np.array([1,2,3,4])
y=np.array([3,5,7,9])

#initial values
m=0 #starts with random guesses
b=0
learning_rate=0.01 #controls how much the model updates each step
epochs=1000 #no. of iterations
n=len(x)
for i in range(epochs):

    #Prediction
    y_pred=m*x+b #equation of line formula
    #gradient
    dm=(-2/n)*np.sum(x*(y-y_pred)) #how much to change slope
    db=(-2/n)*np.sum(y-y_pred) #how much to change intercept

    #update parameter
    m=m-learning_rate*dm
    b=b-learning_rate*db #Move toward Lower error

    print("Slope(m):",m)
    print("Intercept(b):",b)

Slope(m): 0.35000000000000003
Intercept(b): 0.12
```

```
print("Intercept(b):",b)

Slope(m): 0.35000000000000003
Intercept(b): 0.12

[6]: import pandas as pd
from sklearn.linear_model import LinearRegression

#sample data
df=pd.DataFrame({
    'Experience':[1,2,3,4,5],
    'Salary':[30000,35000,40000,45000,50000]
})
#input feature(x)
x=df[['Experience']]

#target Variable(y)
y=df['Salary']

#create model
model=LinearRegression()

#train model (ols happens here)
model.fit(x,y)
predictions=model.predict(x)

#results
print("Slope:",model.coef_[0])
print("Intercept:",model.intercept_)
print("Predictedd Salaries:",predictions)

Slope: 5000.0
Intercept: 25000.0
Predictedd Salaries: [30000. 35000. 40000. 45000. 50000.]

[7]: new_experience=pd.DataFrame({ 'Experience':[6,7,8,9]
})
```

```
[7]: new_experience=pd.DataFrame({'Experience':[6,7,8,9]
    })
```

```
#predict salary for new values
new_predictions=model.predict(new_experience)
print("Preddicted Salaries for 6,7,8,9:")
print(new_predictions)
```

```
Preddicted Salaries for 6,7,8,9:
[55000. 60000. 65000. 70000.]
```

```
[8]: import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))

# Original data points

plt.scatter(df['Experience'], df['Salary'], color='blue', label='Actual Data')

# Regression Line (trained data)

plt.plot(df['Experience'], predictions, color='red', label='Regression Line')

# New predictions

plt.scatter(new_experience['Experience'], new_predictions, color='green')

# Labels

plt.xlabel("Experience (Years)")

plt.ylabel("Salary")

plt.title("Linear Regression: Experience vs Salary")

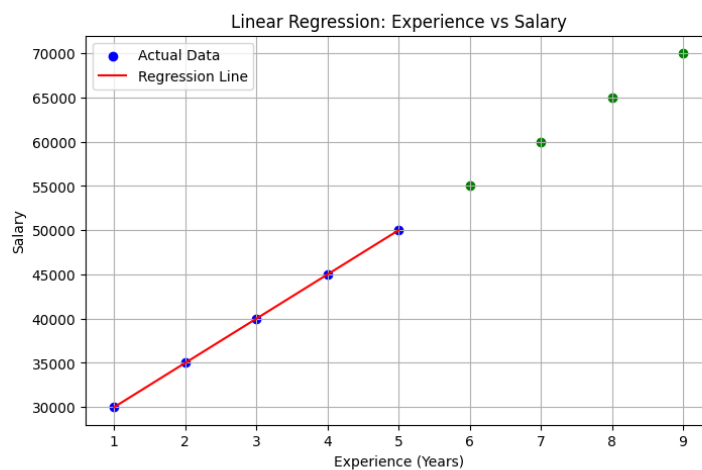
plt.legend()
```

```
plt.title("Linear Regression: Experience vs Salary")

plt.legend()

plt.grid()

plt.show()
```



```
[12]: import pandas as pd
```

Experience (Years)

```
[12]: import pandas as pd
from sklearn.linear_model import LinearRegression

# Sample Data
df = pd.DataFrame({
    'Experience': [1, 2, 3, 4, 5],
    'Salary': [30000, 35000, 40000, 45000, 50000]
})

# Input Feature (X)
X = df[['Experience']]

# Target Variable (y)
y = df['Salary']

# Create Model
model = LinearRegression()

# Train Model (OLS happens here)
model.fit(X, y)

# Predict Salary
predictions = model.predict(X)

# Results
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
print("Predicted Salaries:", predictions)
new_experience = pd.DataFrame({'Experience': [6, 7, 8, 9]
})

# Predict salary for new values
new_predictions = model.predict(new_experience)
print("Predicted Salaries for 6,7,8,9:")
```

```
# Predict salary for new values
new_predictions = model.predict(new_experience)
print("Predicted Salaries for 6,7,8,9:")
print(new_predictions)

Slope: 5000.0
Intercept: 25000.0
Predicted Salaries: [30000. 35000. 40000. 45000. 50000.]
Predicted Salaries for 6,7,8,9:
[55000. 60000. 65000. 70000.]
```

```
[14]: ##Multiple Regression Code
import pandas as pd
from sklearn.linear_model import LinearRegression

# Dataset
df = pd.DataFrame({
    'Area': [4000, 5200, 6500, 4800, 5000],
    'Bedrooms': [5, 4, 6, 3, 4],
    'Age': [12, 5, 1, 4, 2],
    'Price': [800000, 750000, 950000, 500000, 870000]
})

# Features
X = df[['Area', 'Bedrooms', 'Age']] #Using multiple features.

# Target
y = df['Price']

# Model
model = LinearRegression()

# Train
model.fit(X, y) #Learns the relationship between all features and price.

# Prediction
pred = model.predict(X) #predict house price
```

```
model.fit(X,y) #Learn the relationship between the features and price
```

```
# Prediction
pred = model.predict(X) #predict house price
print("Coefficients:", model.coef_)
print("Intercept:", model.intercept_)
```

```
Coefficients: [ -11193.78652722  3999318.15851534 -2222203.42458731]
Intercept: 52497887.82938582
```

```
[15]: #VIF Code
import pandas as pd
from statsmodels.stats.outliers_influence import variance_inflation_factor

# Features
X = df[['Area', 'Bedrooms', 'Age']]

# VIF Calculation
vif = pd.DataFrame()
vif['Feature'] = X.columns
vif['VIF'] = [
    variance_inflation_factor(X.values, i)
    for i in range(X.shape[1])
]
print(vif)
```

```
Feature      VIF
0      Area  29.886901
1  Bedrooms  37.335473
2       Age   2.918224
```

```
[17]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression

# Sample dataset
df = pd.DataFrame({
```

```
from sklearn.linear_model import LinearRegression

# Sample dataset
df = pd.DataFrame({
    'Hours': [1, 2, 3, 4, 5],
    'Marks': [10, 25, 45, 70, 100]
})
X = df[['Hours']]
y = df['Marks']

# Polynomial transformation
poly = PolynomialFeatures(degree=4)
X_poly = poly.fit_transform(X)

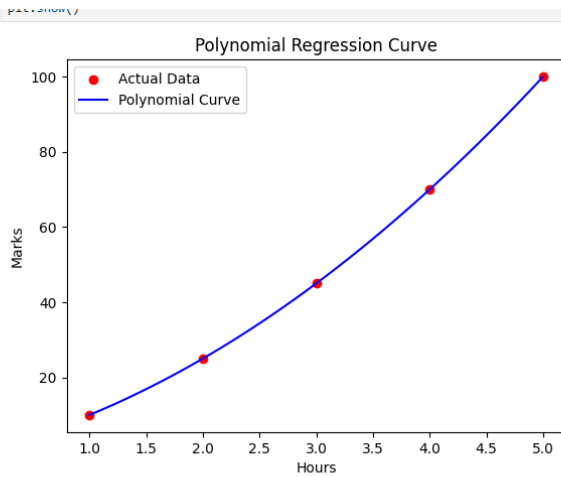
# Model training
model = LinearRegression()
model.fit(X_poly, y)

# Smooth curve generation (IMPORTANT STEP)
X_range = np.linspace(X.min(), X.max(), 100).reshape(-1, 1)
X_range = pd.DataFrame(X_range, columns=['Hours'])
X_range_poly = poly.transform(X_range)

# Predictions for curve
y_curve = model.predict(X_range_poly)

# Plotting
plt.scatter(X, y, color='red', label='Actual Data')
plt.plot(X_range, y_curve, color='blue', label='Polynomial Curve')
plt.title('Polynomial Regression Curve')
plt.xlabel('Hours')
plt.ylabel('Marks')
plt.legend()
plt.show()
```

Polynomial Regression Curve



```
[19]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression

X = np.array([1,2,3,4,5,6,7,8,9,10]).reshape(-1,1)
y = np.array([2,5,8,15,28,35,50,60,85,95])

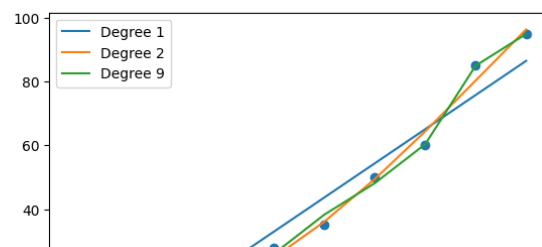
# Degree 1
```

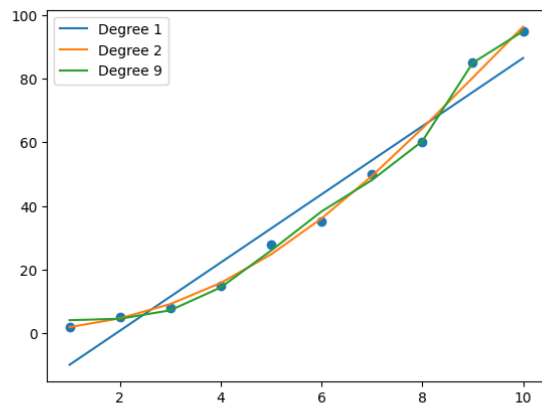
```
y = np.array([2,5,8,15,28,35,50,60,85,95])

# Degree 1
poly1 = PolynomialFeatures(degree=1)
X1 = poly1.fit_transform(X)

# Degree 2
poly4 = PolynomialFeatures(degree=4)
X4 = poly4.fit_transform(X)

# Degree 9
poly9 = PolynomialFeatures(degree=9)
X9 = poly9.fit_transform(X)
model1 = LinearRegression().fit(X1,y)
model2 = LinearRegression().fit(X4,y)
model9 = LinearRegression().fit(X9,y)
plt.scatter(X,y)
plt.plot(X, model1.predict(X1), label="Degree 1")
plt.plot(X, model2.predict(X4), label="Degree 2")
plt.plot(X, model9.predict(X9), label="Degree 9")
plt.legend()
plt.show()
```





```
[20]: ##Ridge Code
from sklearn.linear_model import Ridge
from sklearn.datasets import make_regression

X, y = make_regression(
    n_samples=100,
    n_features=5,
    noise=20,
    random_state=42
)
ridge = Ridge(alpha=1.0)
ridge.fit(X, y)
```

```
X, y = make_regression(
    n_samples=100,
    n_features=5,
    noise=20,
    random_state=42
)
ridge = Ridge(alpha=1.0)
ridge.fit(X, y)
print("Coefficients:")
print(ridge.coef_)
```

```
Coefficients:
[62.4844981  98.07649112  57.01070157  52.31272567  35.35343002]
```

```
[21]: ##Lasso Code
from sklearn.linear_model import Lasso
from sklearn.datasets import make_regression

X, y = make_regression(
    n_samples=100,
    n_features=10,
    noise=20,
    random_state=42
)
lasso = Lasso(alpha=2.0)
lasso.fit(X, y)
print("Coefficients:")
print(lasso.coef_)
```

```
Coefficients:
[17.88753444  54.76076704  0.          60.9632585  90.0800089  68.30461063
 81.35787563  4.03519677  2.86610797  67.99906953]
```

```
[23]: ##ElasticNet Code
from sklearn.linear_model import ElasticNet
from sklearn.datasets import make_regression
```

```
print(lasso.coef_)
```

```
Coefficients:  
[17.88753444 54.76076704  0.          60.9632585  90.0800089  68.30461063  
 81.35787563  4.03519677  2.86610797 67.99906953]
```

```
[23]: ##ElasticNet Code  
from sklearn.linear_model import ElasticNet  
from sklearn.datasets import make_regression
```

```
X, y = make_regression(  
    n_samples=100,  
    n_features=10,  
    noise=20,  
    random_state=42  
)
```

```
elastic = ElasticNet(  
    alpha=3.0,  
    l1_ratio=0.5  
)
```

```
elastic.fit(X, y)  
print("Coefficients:")  
print(elastic.coef_)
```

```
Coefficients:  
[10.65990986 22.10890939  0.49979146 25.9235125  33.08321402 34.26097045  
 35.7929364  -7.32756763  4.65692898 25.41452644]
```

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[ ]:
```